13.Perform Affine Transformation on the given image using python and Open CV

code;

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read the image
img = cv2.imread(r"C:\Users\ROHIT\Pictures\Screenshots\Screenshot 2026-06-17 103005.png")

if img is None:
    print("Image not found!")
else:
    rows, cols = img.shape[:2]

    # Define three points
    pts1 = np.float32([[50, 50], [200, 50], [50, 200]])
    pts2 = np.float32([[10, 100], [200, 50], [100, 250]])

    # Calculate affine transformation
    matrix = cv2.getAffineTransform(pts1, pts2)

    # Apply transformation
    result = cv2.warpAffine(img, matrix, (cols, rows))

    # Display images
    plt.subplot(1, 2, 1)
    plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1, 2, 2)
```

```
plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
plt.title("Affine Transformation")
plt.axis("off")

plt.show() output
```

output:

